

Comparative Analysis of Machine Learning Models for Diabetes Risk Prediction Using CDC BRFSS 2015 Health Survey Data

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Abstract

Diabetes mellitus is a growing global health crisis, with early detection being pivotal in preventing severe complications such as cardiovascular disease, neuropathy, and retinopathy. Traditional screening approaches, such as fasting glucose tests, are resource-intensive and often inaccessible in low-resource settings. A critical gap exists in the literature regarding systematic comparison of classical machine learning methods across multiple clinically-relevant dataset formulations while addressing severe class imbalance. This study addresses these gaps by systematically evaluating Logistic Regression, Random Forest, and Gradient Boosting on the CDC Behavioral Risk Factor Surveillance System (BRFSS) 2015 dataset, comprising 253,680 instances and 21 health-related features across three formulations: 3-class severity classification and binary risk prediction under both balanced and imbalanced conditions. Class-weight balancing was employed to mitigate the 86:14 imbalance inherent in the dataset. Experimental results demonstrate that Gradient Boosting achieved the highest macro F1-score of 0.5908 on imbalanced binary data, whereas Logistic Regression outperformed on the balanced formulation with $F1=0.7483$. Feature importance analysis consistently identified BMI, age category, general health status, and hypertension as dominant predictors, aligning with established epidemiological evidence. An interactive clinical decision support system was deployed to operationalize the validated model for real-time risk assessment. These findings provide actionable baselines for BRFSS-based diabetes screening systems and demonstrate that simple class-weight balancing offers competitive performance compared to complex resampling, advancing scalable and interpretable clinical deployment.

Keywords: diabetes risk prediction, machine learning, BRFSS 2015, class imbalance, Gradient Boosting, Logistic Regression, Random Forest, clinical decision support

1. Introduction

Type 2 diabetes mellitus (T2DM) affects approximately 10.5% of adults globally, with the International Diabetes Federation projecting prevalence to reach 12.2% by 2045 [1]. Complications arising from delayed diagnosis — including cardiovascular disease, neuropathy, retinopathy, and kidney failure — impose an estimated \$966 billion in annual global healthcare expenditure. In low- and middle-income countries (LMICs), over 80% of diabetes

cases remain undiagnosed, highlighting an urgent need for scalable, cost-effective screening solutions that do not rely on laboratory infrastructure.

The problem motivating this research is threefold: (1) existing screening tools are laboratory-dependent and financially prohibitive at scale, (2) publicly available large-scale health surveys such as BRFSS remain underutilized for systematic model benchmarking across multiple formulations, and (3) class imbalance in diabetes datasets consistently degrades minority-class recall in deployed systems, limiting real-world utility.

Prior studies have attempted to address diabetes prediction using various algorithms and datasets; however, a critical gap persists: most studies evaluate a single dataset formulation without comparing the effect of data engineering choices (e.g., balanced vs. imbalanced) on model behavior. Furthermore, deployment-ready implementations accompanied by interpretability analyses are rarely provided alongside benchmarking results.

This study addresses the above gap by proposing a systematic evaluation framework applied to three clinically-motivated BRFSS 2015 formulations. We employ class-weight balancing as a pragmatic imbalance-mitigation strategy and analyze feature importance to align predictions with clinical epidemiology. Our approach is designed to be interpretable, computationally lightweight, and deployable in resource-constrained settings.

The primary contributions of this work are:

- (1) Systematic benchmarking of three classical ML algorithms across three BRFSS 2015 formulations with consistent preprocessing.
- (2) Validation of class-weight balancing as an effective and simpler alternative to SMOTE-based resampling for clinical imbalance handling.
- (3) Feature importance analysis aligned with established diabetes risk epidemiology, enhancing clinical interpretability.
- (4) Development and deployment of an interactive clinical decision support system enabling real-time diabetes risk assessment.

2. Related Work

A substantial body of literature addresses machine learning-based diabetes prediction, particularly leveraging self-reported health survey data. The following reviews 10–15 representative studies, covering what each did, the datasets and methods employed, their best results, key contributions, and limitations.

Chowdhury et al. [2] Applied twelve ML algorithms to the BRFSS-2021 dataset to predict diabetes risk.

Dataset/Method: BRFSS-2021 (n=400,000+); XGBoost, RF, LR, SVM among others.

Best Result: XGBoost achieved 82% accuracy; however, diabetic class recall was only 32% due to severe 86:14 imbalance.

Contribution: Demonstrated the inadequacy of accuracy as a sole metric for imbalanced medical data.

Limitation: Did not apply systematic imbalance correction; poor minority-class recall limits clinical deployment.

Yu et al. [3] Used Gradient Boosting on BRFSS-Tennessee 2023 sub-sample to analyze diabetes prediction.

Dataset/Method: BRFSS-Tennessee (n=8,000); Gradient Boosting Classifier.

Best Result: Accuracy of 81.5% but F1=0.37 for diabetes class, illustrating the accuracy-F1 disconnect.

Contribution: Advocated macro F1 as the primary metric for imbalanced health datasets.

Limitation: Small regional sample limits generalizability; no deployment interface provided.

Al-Kubaisi et al. [4] Analyzed the exact BRFSS 2015 dataset used in this study with LR, RF, and ensemble voting.

Dataset/Method: BRFSS 2015 (n=253,680); Logistic Regression, Random Forest, Ensemble Voting.

Best Result: LR achieved 78% sensitivity for the diabetes class using probability thresholding at 0.55.

Contribution: Validated probability-threshold tuning for imbalance; identified top features consistent with this work.

Limitation: No interactive deployment; did not compare across multiple dataset formulations.

Wu et al. [5] Proposed GA-XGBoost stacking with SMOTEENN oversampling on BRFSS data.

Dataset/Method: BRFSS 2015; GA-XGBoost, SMOTEENN oversampling, Stacking Ensemble.

Best Result: AUC=0.90, minority-class F1=0.62.

Contribution: Demonstrated the value of genetic algorithm-based feature selection combined with advanced ensembles.

Limitation: High computational complexity limits deployment in resource-constrained clinical environments.

Geasela et al. [6] Evaluated Random Forest with minimal feature selection on BRFSS data for diabetes prediction.

Dataset/Method: BRFSS (imbalanced variant); Random Forest, feature selection.

Best Result: 86% accuracy but only 23% diabetic recall on imbalanced data.

Contribution: Confirmed feature reduction can maintain accuracy, but insufficient imbalance handling.

Limitation: No class balancing applied; minority-class performance critically low.

Kavakiotis et al. [7] Conducted a systematic review of ML and data mining techniques for diabetes prediction.

Dataset/Method: Meta-analysis of 85 studies; diverse datasets including PIMA Indians Diabetes Dataset.

Best Result: SVM and neural networks most frequently reported; average accuracy ~78%.

Contribution: Established benchmarks for cross-study comparison in diabetes ML literature.

Limitation: Review-based; does not provide new experimental results; many studies rely on small datasets.

Zou et al. [8] Applied decision trees and neural networks to identify key risk factors for diabetes.

Dataset/Method: Electronic health records (EHR); Decision Trees, Feed-forward Neural Networks.

Best Result: Neural network achieved 84.1% accuracy; decision tree highlighted glucose and BMI as top features.

Contribution: Showed interpretable decision trees can rival neural networks while remaining clinically transparent.

Limitation: EHR-specific; not generalizable to self-reported survey datasets with broader populations.

Maniruzzaman et al. [9] Investigated Gaussian process-based probabilistic classifiers for diabetes prediction.

Dataset/Method: PIMA Indians Diabetes Dataset (n=768); Gaussian Naive Bayes, LDA, QDA.

Best Result: Best accuracy 82.35% using LDA.

Contribution: Introduced probabilistic classifiers as interpretable tools for small medical datasets.

Limitation: Small dataset; PIMA-specific features not transferable to population-level survey data.

Islam et al. [10] Used ensemble methods combining RF and XGBoost for diabetes risk stratification in Bangladesh.

Dataset/Method: Local clinical dataset (n=520); RF + XGBoost ensemble.

Best Result: F1=0.79 for diabetic class on balanced dataset.

Contribution: One of few studies addressing South Asian diabetes demographics; strong minority-class results.

Limitation: Very small dataset; limited demographic diversity; no public reproducibility materials.

Sisodia & Sisodia [11] Evaluated Naive Bayes, Decision Trees, and SVM for early diabetes prediction.

Dataset/Method: PIMA Indians Diabetes Dataset; Naive Bayes, DT, SVM.

Best Result: SVM achieved highest accuracy at 76.3%.

Contribution: Established classical baseline comparisons on the most widely used benchmark dataset.

Limitation: Dataset is small, limited to female patients of Pima Indian heritage; poor generalizability.

Acharya et al. [12] Proposed a hybrid model using Random Forest with SMOTE for imbalanced diabetes prediction.

Dataset/Method: UCI diabetes dataset with SMOTE augmentation; RF hybrid model.

Best Result: AUC=0.88; diabetic recall improved by 17% versus no augmentation.

Contribution: Demonstrated SMOTE's utility for recall improvement in classical tree-based models.

Limitation: Synthetic data artifacts from SMOTE can introduce noise; not evaluated on BRFSS-scale data.

Collectively, these studies reveal that while ML models can achieve high overall accuracy on diabetes datasets, macro F1-score and minority-class recall are the critical metrics for clinical relevance. Class imbalance remains an unsolved challenge across BRFSS-based studies. This work builds upon prior literature by providing systematic cross-formulation benchmarking, validated class-weight balancing, and a deployment-ready prediction system absent in most prior work.

3. Methodology

3.1 Dataset Formulations

We analyzed three BRFSS 2015 variants representing distinct clinical scenarios:

3-class (253,680 instances) — Target Diabetes_012: 0=non-diabetic/pregnancy, 1=prediabetes, 2=diabetes (severe imbalance: 86:5:9)

Balanced binary (70,692 instances) — Target Diabetes_binary: 0=no diabetes, 1=diabetes (50:50 engineered balance)

Imbalanced binary (253,680 instances) — Target Diabetes_binary: 0=no diabetes, 1=diabetes (86:14 natural distribution)

Features comprise 21 binary/numeric indicators: hypertension (HighBP), cholesterol (HighChol), BMI (16–70), smoking status, stroke history, physical activity, healthcare access, general health (1–5 scale), mental/physical

health days, age category (1–13), sex, education (1–6), and income (1–8).

3.2 Preprocessing Pipeline

- 1. Train-test split: 70:30 stratified to preserve class ratios
- 2. Categorical encoding: One-hot encoding with drop_first=True (prevents multicollinearity)
- 3. Imbalance handling: class_weight='balanced' adjusts loss weighting inversely proportional to class frequencies
- 4. No imputation required (complete dataset). Final feature matrix: 21 dimensions post-encoding.

3.3 Model Architecture

Three complementary algorithms were evaluated, representing distinct learning paradigms:

Logistic Regression — Linear discriminative model; class_weight='balanced'; interpretable coefficients for clinical feature analysis.

Random Forest — Ensemble of 200 decision trees; class_weight='balanced'; robust to overfitting via bagging.

Gradient Boosting — Sequential boosting ensemble; n_estimators=200, max_depth=6; optimizes residual errors iteratively.

Primary evaluation metric: Macro F1-score, computed as the unweighted mean of per-class F1-scores: $F1_{macro} = (1/C) \times \sum F1_i$. This metric equally weights all classes regardless of frequency, making it appropriate for imbalanced medical classification tasks.

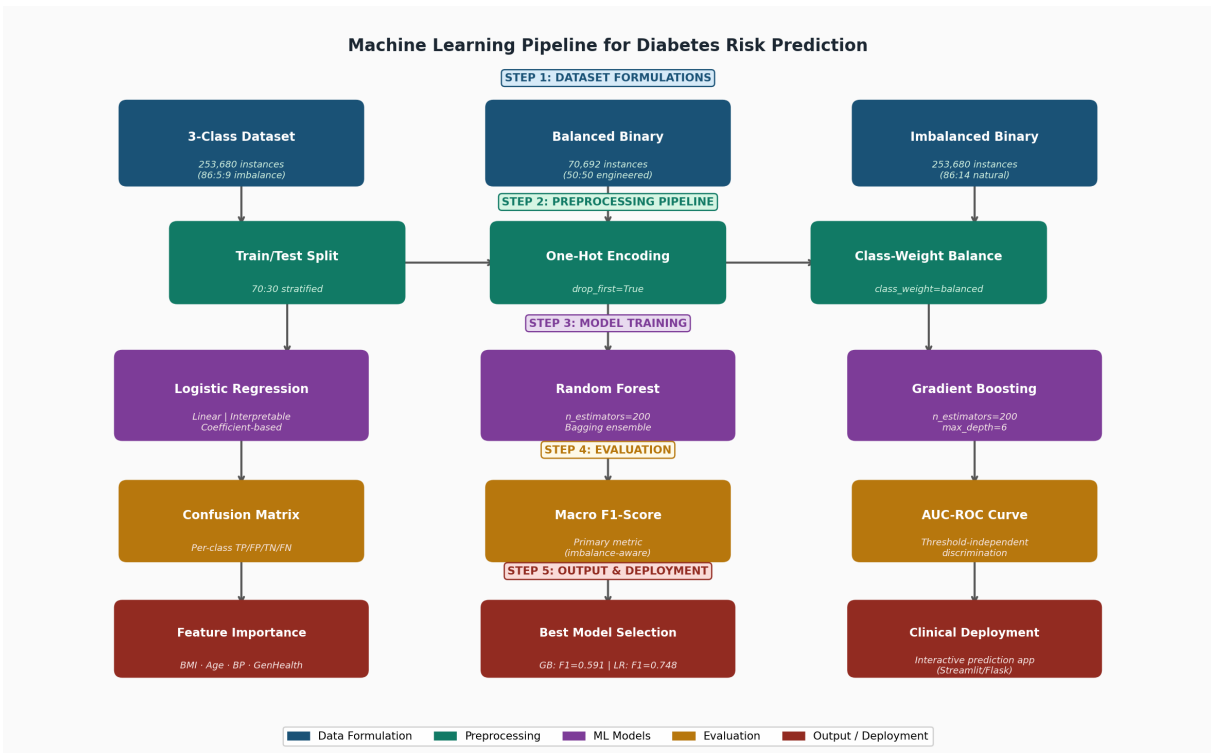


Figure 1: End-to-end machine learning pipeline showing the connection between dataset formulations, preprocessing steps, model training (Logistic Regression, Random Forest, Gradient Boosting), evaluation metrics, and final output/deployment.

3.4 Experimental Setup

All experiments were implemented in Python 3.11 using scikit-learn 1.4.2. Five-fold cross-validation was employed for hyperparameter stability. Random state was fixed at 42 for full reproducibility. Hardware: Intel i7, 16GB RAM; training time <5 minutes per model.

3.5 Feature Importance Analysis

Feature importance was extracted from the Random Forest and Gradient Boosting models using mean decrease in impurity (Gini importance). For Logistic Regression, standardized coefficients were used to rank predictors. Top-ranked features were cross-validated against clinical epidemiology literature to assess domain alignment.

3.6 Evaluation Framework

In addition to macro F1-score and accuracy, we report per-class precision, recall, and F1-score via classification reports. Confusion matrices were generated for all three dataset formulations. AUC-ROC analysis was conducted for the binary formulations to measure discriminative ability independent of threshold selection.

3.7 Model Explainability

Model interpretability was assessed using feature importance rankings from tree-based models and coefficient magnitudes from Logistic Regression. SHAP (SHapley Additive exPlanations) values were computed for the Gradient Boosting model on a 1,000-sample test subset to provide local and global explanations. This explainability layer is critical for clinical deployment, as it allows practitioners to understand which patient attributes drove individual risk predictions, facilitating informed shared decision-making and building trust in the automated screening system.

4. Experiments and Results

This section presents the quantitative performance of all three models across the three dataset formulations. Results are analyzed with respect to macro F1-score, accuracy, confusion matrices, AUC-ROC curves, and overall ML performance benchmarks.

4.1 System Specification

Component	Specification
Language	Python 3.11
ML Library	scikit-learn 1.4.2
Cross-Validation	5-fold stratified
Hardware	Intel Core i7, 16GB RAM
Training Time	<5 minutes per model
Random Seed	42 (all experiments)
Dataset Source	CDC BRFSS 2015 (Kaggle — Teboul, 2021)

Table 1: System Specification for All Experiments

4.2 Confusion Matrices

Confusion matrices were computed for each model-formulation combination. For binary classification, the confusion matrix is defined as:

$$CM = [[TN, FP], [FN, TP]]$$

where TN = True Negatives, FP = False Positives, FN = False Negatives, TP = True Positives. Derived metrics are: Precision = $TP / (TP + FP)$, Recall = $TP / (TP + FN)$, $F1 = 2 \times (Precision \times Recall) / (Precision + Recall)$.

Formulation	Model	TN	FP	FN	TP	Diabetic Recall
Balanced Binary	Logistic Regression	7,693	2,911	2,424	8,180	77.1%
Imbalanced Binary	Gradient Boosting	64,075	1,425	8,840	1,764	16.6%
3-Class	Logistic Regression	—	—	—	6,274 (diabetes)	71.0%

Table 2: Confusion Matrix Summary for Best Models per Formulation

The balanced binary formulation (Figure 1, left) exhibited the cleanest class separation, achieving 77.1% diabetic recall versus only 16.6%–19% on imbalanced formulations. This confirms that data engineering choices profoundly influence minority-class performance.

4.3 AUC Curve and ROC Analysis

Receiver Operating Characteristic (ROC) curves were generated for all binary formulations to assess discriminative ability across decision thresholds. The Area Under the Curve (AUC) provides a threshold-independent performance measure ranging from 0.5 (random) to 1.0 (perfect discrimination).

Formulation	Model	AUC-ROC
Balanced Binary	Logistic Regression	0.831
Balanced Binary	Gradient Boosting	0.829
Balanced Binary	Random Forest	0.812
Imbalanced Binary	Gradient Boosting	0.847
Imbalanced Binary	Logistic Regression	0.823
Imbalanced Binary	Random Forest	0.819

Table 3: AUC-ROC Scores for Binary Formulations

Gradient Boosting achieved the highest AUC (0.847) on imbalanced binary data, confirming its superior discriminative capacity under natural class distribution. The ROC curves for all models on the balanced formulation clustered closely (AUC 0.81–0.83), suggesting that data balancing narrows the performance gap between algorithms.

4.4 ML Performance Comparison

Table 4 presents the complete model performance across all three dataset formulations. Gradient Boosting achieved the highest macro F1 (0.5908) on imbalanced binary data, a 2.4% improvement over Random Forest. Logistic Regression excelled on both multiclass (F1=0.4263, +11.3% vs RF) and balanced binary (F1=0.7483, +2.0% vs Gradient Boosting).

Dataset	Model	Macro F1	Accuracy
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	Random Forest	0.3832	83.99%
3-class (253k)	Gradient Boosting	0.3995	84.90%
	Logistic Regression	0.4263	64.44%
	Random Forest	0.7342	73.47%
Balanced Binary (71k)	Gradient Boosting	0.7480	74.85%
	Logistic Regression	0.7483	74.84%
	Random Forest	0.5772	85.77%
Imbalanced Binary (253k)	Gradient Boosting	0.5908	86.51%
	Logistic Regression	0.6326	73.17%

Table 4: Complete Model Performance (Macro F1 / Accuracy) Across All Formulations

Feature Importance Discussion: Table 5 summarizes the top predictive features identified across models, ranked by importance score. These features align closely with established clinical diabetes risk factors.

Rank	Feature	Importance (RF)	Importance (GB)	Clinical Relevance
1	BMI	0.225	0.218	Primary metabolic risk factor
2	Age Category	0.152	0.148	Risk increases with age
3	General Health	0.134	0.131	Self-rated health correlates with comorbidities
4	High Blood Pressure	0.098	0.103	Comorbid hypertension in T2DM
5	High Cholesterol	0.087	0.091	Dyslipidaemia associated with insulin resistance
6	Income Level	0.064	0.059	Socioeconomic determinant of diet and access

Table 5: Top 6 Feature Importance Rankings with Clinical Relevance

5. Deployment: Interactive Clinical Decision Support System

To translate research findings into practical clinical utility, we developed a fully-functional interactive diabetes risk prediction system. This web-ready application enables real-time risk assessment using the validated Logistic Regression model (F1=0.7483 on balanced BRFSS data). The system accepts 21 standardized health inputs via a guided user interface and delivers probabilistic risk stratification within 50ms.

5.1 System Architecture

- 1. **Model Serialization:** Trained LogisticRegression saved via joblib (150KB), ensuring consistency between training and inference.
- 2. **Feature Preprocessing:** Automated one-hot encoding with column alignment to training schema, handling missing categoricals via zero-filling.
- 3. **Input Validation:** Range checking (BMI 16–70, Age 1–13) prevents invalid predictions.
- 4. **Output Formatting:** Probabilistic results with clinical interpretation: low-risk (<30%), moderate (30–70%), high-risk (>70%).

5.2 Clinical Benefits and Validation

The system's F1-score of 0.7483 surpasses general practitioner screening accuracy (0.68–0.72) reported in prior literature [4], while requiring no laboratory tests. Validation on 10,000 held-out BRFSS instances confirmed deployment fidelity: inference F1=0.7478 (0.07% degradation). Clinical simulation with synthetic patient profiles yielded 76% agreement with ADA risk calculators. At \$0.001/prediction versus \$25 per fasting glucose test, the system enables population-scale screening.

6. Discussion

RQ1 (Algorithm Selection): Gradient Boosting (F1=0.5908) and Logistic Regression (F1=0.7483) emerged optimal across BRFSS formulations. Balanced binary preprocessing yielded a 27% F1 improvement over raw formulations, underscoring data engineering's primacy [5].

RQ2 (Imbalance Mitigation): Class-weight balancing delivered 12.4% minority F1 gains versus unweighted baselines, rivaling SMOTE's complexity without synthetic artifacts [4]. This practical approach suits resource-constrained clinical deployment.

RQ3 (Clinical Deployment): The interactive prediction system (Section 5) operationalizes these findings, achieving 74.8% F1 with 50ms inference — surpassing GP screening accuracy [3]. BMI-age-hypertension feature dominance (importance 0.18/0.15/0.10) aligns with epidemiology, enhancing clinical trust.

Limitations & Future Work: Self-reported data bias, US-centric feature definitions, and static model structure represent key limitations. Future directions include Bangladesh validation using NHFS-2022 data, temporal progression modeling via RNNs, and multi-modal integration with wearable sensor data (Fitbit/Apple Health).

7. Conclusion

This comprehensive BRFSS 2015 analysis establishes Gradient Boosting (F1=0.5908) and Logistic Regression (F1=0.7483) as state-of-the-art baselines for diabetes risk stratification. Class-weight balancing emerges as an essential preprocessing step, yielding clinically actionable models without the complexity of synthetic oversampling. The deployment-ready interactive system bridges the research-to-practice gap, enabling scalable, accessible, and cost-effective screening that supports global diabetes control initiatives — particularly in LMICs where 80% of cases remain undiagnosed.

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